

claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf_28916f46-0f0\agent-ae93ff82195d06de4.jsonl`  
- SHA-256: `a0e8a367efc6a5ec88370e4849470c88360c114cf59bf7185525f5772a5f819f`  
- Source modified: `2026-05-30T08:41:00+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_28916f46-0f0`  
- session_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`
```

Transcript

1. user (2026-05-30T08:37:42.477Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

RESEARCH GOAL: Produce a decision-grade shortlist of data-annotation VENDORS to build "golden set" ground-truth labels for a badminton highlights product, with a STRONG PREFERENCE for vendors based in or operating from Bangalore / India (the founder is in Bangalore and prefers local vendors for easier coordination, possible site visits, and data residency). The annotation task is TEMPORAL VIDEO SEGMENT annotation: marking the [start, end) time ranges of each "rally" in badminton match videos ? i.e. event / temporal-action labeling on video, NOT just image bounding boxes. Output is a simple per-video timestamp list (CSV of start,end). Volume: a constantly growing collection of licensed/owned match videos (no end-user uploads).

For EACH candidate vendor, verify against PRIMARY sources (the vendor's own website/docs) and clearly FLAG anything unverifiable (pricing is usually quote-only):

1. Company & location ? HQ + delivery centers; confirm Bangalore / India presence; flag if purely overseas.
2. Capability ? do they do VIDEO temporal/event annotation (timestamps/segments) or video tracking, or only static image labeling? Sports / media-video experience is a plus.
3. Pricing model ? per video-minute / per task / per annotator-hour / managed-team; any public rates; minimum engagement; willingness to run a small paid PILOT.
4. Data security & compliance ? ISO 27001, SOC 2, GDPR, and especially India's DPDP Act 2023; willingness to sign a DPA; "no training/reuse of our data"; delete-on-delivery; access controls; secure-facility / VDI options.
5. Delivery & integration ? programmatic API for submit/fetch? turnaround SLAs? can they support BOTH a fast tier (hours) and a batch tier (days)?
6. Quality process ? multi-annotator consensus, gold-standard/honeypot checks, inter-annotator-agreement (IAA) reporting, QA review layers.

Candidate vendors to investigate and VERIFY (do not assume ? confirm or correct each, and find others, especially Bangalore-local): iMerit, Playment (and whether TELUS International acquired it), Cogito Tech, Anolytics, Infolks, Learning Spiral, Indika AI, Tech Mahindra, Sama, Scale AI, Labelbox, Toloka, SuperAnnotate, Encord, CloudFactory. Also identify any Bangalore-based startups specializing in video annotation.

Also briefly cover: (a) typical market rate RANGE for temporal video annotation in India vs US/global; and (b) the AUTOMATED-annotation alternative ? could a hosted VLM (e.g. Gemini) or an open model serve as a "fast tier" oracle for surprise regression tests, and what that implies (it must be a DIFFERENT model lineage than the production tool to avoid shared-blind-spot bias).

DELIVERABLE: a ranked shortlist (favoring Bangalore/India) with a comparison table (vendor | location | video-temporal capability | pricing model | security/DPA & DPDP posture |

API/turnaround | QA process | pilot availability | verdict), plus 2?3 recommended vendors to approach for a pilot bake-off, and explicit flags on every unverifiable claim (especially pricing). Prioritize accuracy and primary-source citations over breadth.

Claim under review

"Sama DOES perform temporal video annotation suitable for marking time-based events: its site lists 'Object Tracking and Temporal Attributes,' uses 'linear interpolation between frames,' supports 'static and dynamic annotations' with 'complex object changes over time,' and explicitly cites 'player tracking for sports analytics' as a use case ? directly relevant to badminton rally segmentation, though no detailed sports portfolio is shown."

****Source:**** <https://www.sama.com/> (primary)

****Supporting quote:**** "Object Tracking and Temporal Attributes ... Utilize linear interpolation between frames ... player tracking for sports analytics"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-30T08:37:47.076Z)

I'll verify this claim about Sama's temporal video annotation capabilities by checking primary sources.

3. user (2026-05-30T08:38:06.491Z)

Video Annotation Mentions Found

The webpage contains the following reference to video annotation capabilities:

****Exact quote:****

"Video Annotation - Object Tracking and Temporal Attributes"

This appears in the navigation menu under their Data Annotation services section. The company offers video annotation specifically designed for object tracking and capturing temporal attributes, though the page does not elaborate further on technical methodologies like frame interpolation or specific sports analytics applications.

4. user (2026-05-30T08:38:16.062Z)

Web search results for query: "Sama.com "Object Tracking and Temporal Attributes" "linear interpolation between frames" video annotation"

Links: [{"title":"Image Annotation Services for Computer Vision Model Training | Sama","url":"https://www.sama.com/2d-image-annotation-services"}, {"title":"Video Annotation Services: Model Training Support & Outsourcing | Sama","url":"https://www.sama.com/2d-video-annotation-tool-services"}, {"title":"Object Tracking with Frame Level-Labeling | Sama","url":"https://www.sama.com/blog/object-tracking-in-samahub-with-frame-level-labeling"}, {"title":"Data Annotation Guide 2026: Complete Overview & Downloadable Checklist | Sama","url":"https://www.sama.com/blog/data-annotation-guide"}, {"title":"Video Annotation -

```
Sama", "url": "https://docs.sama.com/video-annotation/video-annotation/version/3?kb_language=en_US"}, {"title": "Efficient video annotation with visual interpolation and frame selection guidance", "url": "https://arxiv.org/pdf/2012.12554"}, {"title": "Annotating general objects in video", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9118886"}, {"title": "Semi-Automatic Annotation For Visual Object Tracking", "url": "https://arxiv.org/pdf/2101.06977"}]
```

Based on the search results, I found relevant information about Sama.com's video annotation capabilities related to object tracking and linear interpolation.

Sama offers the ability to create custom classes, multiple attributes per instance, and hierarchical relationships, while powering object tracking with ML Assist for faster high-quality annotations and using ML-assisted or linear interpolation between frames.

Sama supports dynamic labeling for object characteristics that change over time, such as visibility percentage of an object or a specific set of characteristics like emotions. Additionally, Sama's built-in automated interpolation between video frames helps ensure efficient, high quality labeled training data for a variety of object tracking use cases.

The platform also supports frame-level labeling technology that allows unique objects to be dynamically tracked across a video, with changes in position and pose captured using annotation shapes like cuboids, polygons and bounding boxes.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-30T08:38:16.468Z)

Web search results for query: "Sama "Object Tracking and Temporal Attributes" "linear interpolation between frames" video annotation"

```
Links: [{"title": "Image Annotation Services for Computer Vision Model Training | Sama", "url": "https://www.sama.com/2d-image-annotation-services"}, {"title": "Video Annotation Services: Model Training Support & Outsourcing | Sama", "url": "https://www.sama.com/2d-video-annotation-tool-services"}, {"title": "Object Tracking with Frame Level-Labeling | Sama", "url": "https://www.sama.com/blog/object-tracking-in-samahub-with-frame-level-labeling"}, {"title": "Temporal Consistency in Video Annotation | Keylabs", "url": "https://keylabs.ai/blog/temporal-consistency-in-video-annotation/"}, {"title": "Video Annotation - Sama", "url": "https://docs.sama.com/video-annotation/video-annotation/version/3?kb_language=en_US"}, {"title": "Efficient video annotation with visual interpolation and frame selection guidance", "url": "https://arxiv.org/pdf/2012.12554"}, {"title": "SAM 3 for Video: Concept-Aware Segmentation and Object Tracking - PyImageSearch", "url": "https://pyimagesearch.com/2026/03/02/sam-3-for-video-concept-aware-segmentation-and-object-tracking/"}, {"title": "CVAT Accelerates Video Annotation with SAM 2 Integration | CVAT Changelog", "url": "https://www.cvat.ai/resources/changelog/video-annotation-sam-2"}]
```

Based on the search results, I found relevant information about Sama's video annotation capabilities related to object tracking and temporal attributes with linear interpolation between frames.

Sama allows users to create custom classes, multiple attributes per instance, and hierarchical relationships while powering object tracking with ML Assist for faster high-quality annotations, and using ML-assisted or linear interpolation between frames.

Additionally, Sama supports dynamic labeling for object characteristics that change over time, such as visibility percentage of an object or specific sets of characteristics, and Sama's built-in automated interpolation between video frames helps ensure efficient, high-quality labeled training data for a variety of object tracking use cases.

Their frame-level labeling technology allows unique objects to be dynamically tracked across a video, with changes in position and pose captured with annotation shapes like cuboids, polygons

and bounding boxes.

These features align with what you searched for regarding object tracking, temporal attributes, and linear interpolation in video annotation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. user (2026-05-30T08:38:17.516Z)

Web search results for query: "Sama video annotation "object tracking" temporal "sports analytics" player tracking"

Links: [{"title": "Video Annotation Services: Model Training Support & Outsourcing | Sama", "url": "https://www.sama.com/2d-video-annotation-tool-services"}, {"title": "Video Annotation for Computer Vision [A Practical Guide, Techniques & Best Practices] | Sama", "url": "https://www.sama.com/blog/video-annotation-for-computer-vision"}, {"title": "Video Annotation - Sama", "url": "https://docs.sama.com/video-annotation/video-annotation/version/3?kb_language=en_US"}, {"title": "SAM 2 Object Tracking - Labellerr Documentation", "url": "https://docs.labellerr.com/features/sam-2-object-tracking"}, {"title": "Team-Aware Football Player Tracking with SAM:", "url": "https://arxiv.org/pdf/2512.08467"}, {"title": "Team-Aware Football Player Tracking with SAM: An Appearance-Based Approach to Occlusion Recovery", "url": "https://arxiv.org/html/2512.08467"}, {"title": "SAM 3: Video to Video Segmentation + Object Tracking | fal", "url": "https://fal.ai/models/fal-ai/sam-3/video"}, {"title": "Towards long-term player tracking with graph hierarchies and domain-specific features", "url": "https://arxiv.org/pdf/2502.21242"}, {"title": "Spatio-Temporal Analysis of Team Sports -- A Survey", "url": "https://arxiv.org/pdf/1602.06994"}]

I found relevant search results for your query. Here's what the results show:

Sama's Video Annotation Services

Sama offers video annotation services that assist in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare.

The company has expertise in handling vast volumes of video data and ensures accurate and high-quality annotations, with skilled annotators meticulously labeling objects, actions, and events to improve computer vision models' performance.

Temporal Aspects and Video Annotation

Video annotation is the better choice when motion, behavior, or temporal patterns matter, such as in autonomous driving, sports analytics, or safety monitoring. Unlike static image annotation, video annotation considers temporal continuity, ensuring that annotations are consistent and accurate across consecutive frames, which makes it particularly useful for applications that require tracking movement or changes over time.

Object Tracking Techniques

Video annotation processes may include techniques like bounding boxes, polygons, semantic segmentation, or keypoint tracking. Annotators can streamline their workload by marking only the keyframes, which represent meaningful points of change, and the model can then generate the movement between those points through interpolation.

The search results demonstrate that Sama provides comprehensive video annotation services specifically suited for sports analytics applications involving object and player tracking with temporal consistency.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-30T08:38:21.389Z)

Search Results

I found one relevant mention:

****Object Tracking and Temporal Attributes:****

The page includes a navigation link to video annotation services that states: "Video Annotation - Object Tracking and Temporal Attributes"

This is the only explicit reference to object tracking or temporal attributes in the provided content. There are no mentions of linear interpolation, static/dynamic annotations, complex object changes, sports analytics, or player tracking.

8. user (2026-05-30T08:38:29.297Z)

Web search results for query: "Sama.com video annotation services capabilities temporal"

Links: [{"title":"Video Annotation - Sama","url":"https://docs.sama.com/en_US/annotation-products/video-annotation"}, {"title":"Functions by annotation product - Sama","url":"https://docs.sama.com/functions-by-annotation-product"}, {"title":"Data Annotation Services for AI & ML Models | Sama","url":"https://www.sama.com/data-annotation-solution"}, {"title":"Video Annotation Services: Model Training Support & Outsourcing | Sama","url":"https://www.sama.com/2d-video-annotation-tool-services"}, {"title":"Video Annotation for Computer Vision [A Practical Guide, Techniques & Best Practices] | Sama","url":"https://www.sama.com/blog/video-annotation-for-computer-vision"}, {"title":"Data Annotation Guide 2026: Complete Overview & Downloadable Checklist | Sama","url":"https://www.sama.com/blog/data-annotation-guide"}, {"title":"Scematics: The Ultimate Image & Video Annotation Tool","url":"https://scematics.io/pages/resource/blogs/SAM3-The-Image-Annotation-Model"}]

Based on the search results, here's what I found about Sama.com's video annotation services and their temporal capabilities:

Temporal Capabilities

Sama's video annotation considers temporal continuity, ensuring that annotations are consistent and accurate across consecutive frames. Sama video annotation experts precisely track temporal and spatial features of target objects and improve machine learning model performance.

Key Annotation Techniques

The process may include techniques like bounding boxes, polygons, semantic segmentation, or keypoint tracking.

Service Approach

Each project is assigned to a dedicated team at Sama consisting of internal experts with deep industry knowledge, and these teams undergo project-specific training for video annotation services. Expert annotators leverage the ML-powered platform with a multi-level quality management process coupled with Auto QA features to catch errors and edge cases early.

Quality Assurance

Sama consistently has a 99% client acceptance rate no matter how complex workflows are.

Use Cases

Applications include player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-05-30T08:38:28.255Z)

Web search results for query: ""sama" "static and dynamic annotations" "complex object changes over time""

Links: [{"title": "Video Annotation Services: Model Training Support & Outsourcing | Sama", "url": "https://www.sama.com/2d-video-annotation-tool-services"}, {"title": "Objects in the workspace - Sama", "url": "https://docs.sama.com/annotation-products/objects-in-the-workspace/version/12?kb_language=en_US"}, {"title": "Sama Launches First Scalable Medium-Length Sequence Annotation Solution for Automotive AI | Sama", "url": "https://www.sama.com/blog/sama-launches-first-scalable-medium-sequence-annotation-solution-for-automotive-ai"}, {"title": "Data Annotation & Labeling Company | Sama", "url": "https://www.sama.com/"}, {"title": "Scala Saturday: Static vs dynamic typing | by Mark Canlas | Medium", "url": "https://medium.com/@markcanlasnyc/scala-saturday-static-vs-dynamic-typing-2bbe6a283233"}, {"title": "9 Using Dynamic JAXB", "url": "https://docs.oracle.com/middleware/1212/toplink/TLJAX/dynamic_jaxb.htm"}, {"title": "How to dynamically transform AWS SAM parameters into a new variable for reuse", "url": "https://blowstack.com/blog/how-to-dynamically-transform-aws-sam-parameters-into-a-new-variable-for-reuse"}]

Based on the search results, I found relevant information about Sama's approach to static and dynamic annotations for tracking complex object changes over time.

Sama helps improve ML model accuracy by tracking complex object changes over time with static and dynamic annotations.

In terms of implementation, the Class of an object defines the static and dynamic attributes the object and its shapes will have. This approach allows for comprehensive annotation of objects as they change throughout video sequences.

Sama has demonstrated significant capability in handling this type of complex annotation work, particularly in automotive applications. For one complex 3D workflow, Sama delivered 10 million shapes per day with a 99% quality rate for both 3D dynamic and static objects.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-05-30T08:38:26.794Z)

Search Results for Video Annotation Features

I found one relevant mention in the navigation menu:

Object Tracking and Temporal Attributes

This appears as a subtitle under the "Video Annotation" service offering in Sama's main navigation menu. However, the page content does not include any detailed discussion of linear interpolation, static/dynamic annotations, complex object changes, sports analytics, or player tracking applications.

11. user (2026-05-30T08:38:34.389Z)

MCP error -32602: Input validation error: Invalid arguments for tool navigate: [

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{
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  "expected": "number",
  "received": "undefined",
  "path": [
    "tabId"
  ],
  "message": "Required"
```

```
}  
]
```

12. user (2026-05-30T08:38:47.673Z)

Sama's Annotation Capabilities and Use Cases

Annotation Capabilities Mentioned:

1. **Image Annotation**: "Object Detection, Localization and Segmentation"
2. **Video Annotation**: "Object Tracking and Temporal Attributes"
3. **3D Point Cloud Annotation**: "Lidar, Radar, and 3D Data Labelling"
4. **Text Annotation**: "Instruction following, preference ranking, and factuality"

Use Cases Listed:

- ADAS & Autonomous Vehicles
- Banking, Financial Services, & Insurance
- Consumer & Media
- Food & Agriculture
- Generative AI
- Retail & E-Commerce
- Robotics & Manufacturing

Notable Findings:

The page explicitly mentions **"Object Tracking and Temporal Attributes"** under Video Annotation services, though no additional details about implementation methods (such as linear interpolation, frame-by-frame processing, or specific tracking techniques) are provided.

No references found to sports analytics, player tracking, static/dynamic annotations, or complex object change management on this page.

13. user (2026-05-30T08:38:49.285Z)

Analysis of Page Content

"Sports analytics" and "player tracking": These phrases do not appear anywhere on the page.

"Static and dynamic annotations": This phrase does not appear anywhere on the page.

Use Cases and Industries Named:

Industries:

- ADAS & Autonomous Vehicles
- Banking, Financial Services, & Insurance (BFSI)
- Consumer & Media
- Food & Agriculture
- Generative AI
- Retail & E-Commerce
- Robotics & Manufacturing

Specific Use Cases/Applications:

- Object Detection, Localization, and Segmentation (Image)
- Object Tracking and Temporal Attributes (Video)
- Lidar, Radar, and 3D Data Labeling
- Instruction following, preference ranking, and factuality (Text)
- Model Evaluation
- Data Validation
- Product image annotation and attributes (Walmart example)
- Waste object labeling (PolyPerception)

- 3D foot scan annotations (Volumental)
- Object detection for security footage (Indoor Robotics)

14. user (2026-05-30T08:39:01.236Z)

Web search results for query: "site:sama.com sports analytics player tracking"

Links: [{"title": "Video Annotation Services: Model Training Support & Outsourcing | Sama", "url": "https://www.sama.com/2d-video-annotation-tool-services"}, {"title": "Key Point Annotation | Sama", "url": "https://www.sama.com/keypoint-annotation-tool"}, {"title": "Data Annotation Services for AI & ML Models | Sama", "url": "https://www.sama.com/data-annotation-solution"}, {"title": "Data Annotation & Labeling Company | Sama", "url": "https://www.sama.com/"}, {"title": "Sama Curate | Data Curation Tools for AI & ML Models", "url": "https://www.sama.com/data-curation-tools-ai-ml-models"}, {"title": "Our Impact | Sama", "url": "https://www.sama.com/impact"}, {"title": "ML Assisted Annotation Powered by MICROMODEL Technology | Sama", "url": "https://www.sama.com/blog/ml-assisted-annotation-micromodels"}, {"title": "Data Annotation Experts at Scale | Sama", "url": "https://www.sama.com/talk-to-an-expert"}, {"title": "Why Choose Sama", "url": "https://www.sama.com/why-sama"}, {"title": "Breaking Down The Scale AI Acquisition and How Sama Is Different | Sama", "url": "https://www.sama.com/blog/breaking-down-the-scale-ai-acquisition-and-how-sama-is-different"}]

Based on the search results, here's what I found about Sama's sports analytics player tracking services:

Sama's latest technology assists in player tracking for sports analytics through their comprehensive video annotation capabilities. The company offers several tools and services relevant to this application:

****Video Annotation Services****: Sama has expertise in handling vast volumes of video data, with skilled annotators who meticulously label objects, actions, and events, improving computer vision models' performance.

****Keypoint Annotation Tools****: Keypoint annotation tools precisely detect shape variations for motion tracking, facial landmark detection, and hand gesture recognition, and sports analytics is listed as a use case.

****Quality and Tracking Capabilities****: Sama reliably delivers high quality annotated data using advanced object detection, segmentation, and tracking capabilities for complex requirements.

The company leverages a combination of human expertise and AI-powered solutions to support sports analytics applications that require accurate player tracking and movement analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. user (2026-05-30T08:38:58.234Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

16. user (2026-05-30T08:39:02.135Z)

Web search results for query: "Sama "player tracking for sports analytics" video annotation use case"

Links: [{"title": "Video Annotation Services: Model Training Support & Outsourcing | Sama", "url": "https://www.sama.com/2d-video-annotation-tool-services"}, {"title": "SAM 2 Object Tracking - Labellerr Documentation", "url": "https://docs.labellerr.com/features/sam-2-object-tracking"}, {"title": "Annotating Sports Footage for Player Tracking AI", "url": "https://datavlab.ai/post/how-to-annotate-sports-footage-for-player-tracking-"}]

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ai"}, {"title": "Team-Aware Football Player Tracking with SAM", "url": "https://arxiv.org/pdf/2512.08467"}, {"title": "Generate Faster Annotations For Sports Analysis With Labellerr Using SAM", "url": "https://www.labellerr.com/blog/transforming-sports-analysis-with-player-segmentation-using-labellerr/"}, {"title": "Video Annotation - Sama", "url": "https://docs.sama.com/video-annotation/video-annotation/version/3?kb_language=en_US"}, {"title": "DataVLab | Computer Vision Sports Analytics", "url": "https://datavlab.ai/post/computer-vision-sports-analytics"}, {"title": "Sports Data Tagging & Annotation, Image & Video | Pixel Annotation", "url": "https://pixelannotation.com/sports-ai-data-annotation/"}, {"title": "Video Annotation for Computer Vision [A Practical Guide, Techniques & Best Practices] | Sama", "url": "https://www.sama.com/blog/video-annotation-for-computer-vision"}, {"title": "YOLO11 vs SAM3: AI-Based Basketball Analytics System", "url": "https://www.labellerr.com/blog/yolo11-vs-sam3-basketball/"}]
```

Based on the search results, here's what I found about Sama's player tracking for sports analytics use case:

Sama's Player Tracking for Sports Analytics

Sama assists with player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare.

Video Annotation Services

Sama handles vast volumes of video data with accurate and high-quality annotations, where skilled annotators meticulously label objects, actions, and events, improving computer vision models' performance.

Technical Capabilities

Sama's video annotation services include several techniques:

- Lines and arrows are used in 2D video annotation to highlight important features in a video for object tracking or scene understanding.
- Bounding boxes, polygons, keypoints & points, lines & arrows, and cuboids support a wide range of use cases from ADAS & autonomous vehicles to video analytics for e-commerce and safety surveillance.

Quality Assurance

Sama continually enhances its video annotation solutions through iterative feedback and utilizes level feedback, percentage of data set review, and automated scoring, bolstered by real-time analytics that offer immediate actionable insights.

The platform enables sports organizations to leverage high-quality annotated video data for training machine learning models that can automatically track players and extract actionable insights from sports footage.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-05-30T08:39:09.690Z)

Web search results for query: "sama.com sports analytics player tracking annotation"

Links: [{"title": "Hitting Revenues Out of the Park with Accurate Data for Sports Analytics | Sama", "url": "https://www.sama.com/blog/data-labeling-challenges-for-sports-analytics"}, {"title": "Generate Faster Annotations For Sports Analysis With Labellerr Using SAM", "url": "https://www.labellerr.com/blog/transforming-sports-analysis-with-player-segmentation-using-labellerr/"}, {"title": "SAM 2 Object Tracking - Labellerr Documentation", "url": "https://docs.labellerr.com/features/sam-2-object-tracking"}, {"title": "Team-Aware Football Player Tracking with SAM", "url": "https://arxiv.org/pdf/2512.08467"}, {"title": "YOLO11 vs SAM3: AI-Based Basketball Analytics System", "url": "https://www.labellerr.com/blog/yolo11-vs-"}]

```
sam3-basketball/"},"title":"Team-Aware Football Player Tracking with SAM: An Appearance-Based Approach to Occlusion Recovery","url":"https://arxiv.org/html/2512.08467"},"title":"Video Annotation for Computer Vision [A Practical Guide, Techniques & Best Practices] | Sama","url":"https://www.sama.com/blog/video-annotation-for-computer-vision"},"title":"Using SAM for the Prediction of Segmentations on the Kaggle Football Player Segmentation Dataset - Voxel51","url":"https://voxel51.com/blog/computer-vision-sam-for-prediction-kaggle-football-player-segmentation-dataset"},"title":"Annotating Sports Footage for Player Tracking AI","url":"https://datavlab.ai/post/how-to-annotate-sports-footage-for-player-tracking-ai"},"title":"Data Annotation Services for AI & ML Models | Sama","url":"https://www.sama.com/data-annotation-solution"}]]
```

Based on the search results, here's what I found about Sama.com's sports analytics player tracking annotation services:

Sama's Sports Analytics Solution

Sama provides ML Professionals and AI team Leads with an indispensable solution for Computer Vision data labeling. The platform specifically addresses the challenges of sports analytics data annotation.

Key Capabilities

Sama handles the challenge of labeling videos and images in varying lighting and weather conditions, as well as close interactions and equipment that result in partial or full occlusion of players or the ball.

Sama reliably delivers high quality annotated data using advanced object detection, segmentation, and tracking capabilities for complex requirements.

Service Quality

Sama consistently has a 99% client acceptance rate regardless of workflow complexity. Expert annotators leverage the ML-powered platform with a multi-level quality management process coupled with Auto QA features to catch errors and edge cases early.

Sports-Specific Features

Camera feeds from previous matches and data on body angles when pitching or kicking collected from players' wearables all funnel into algorithms that analyze large banks of data to understand what it takes for a win. The platform enables teams to use these insights to contrast their performance and improve their approach.

Sama's expertise in sports analytics annotation comes from years of experience handling the unique complexities of video annotation in athletic settings, including player occlusion, varying lighting conditions, and real-time motion tracking requirements.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. user (2026-05-30T08:39:15.674Z)

Search Results

I searched the entire page text for the words "sport," "sports," "player," "athlete," "game," and "match."

****Result: None of these words appear anywhere on the page.****

The content focuses exclusively on Sama's data annotation and labeling services for AI development, covering topics such as computer vision, autonomous vehicles, retail, agriculture, and generative AI?with no references to sports, athletes, players, or related activities.

19. user (2026-05-30T08:39:25.568Z)

Structured output provided successfully

20. assistant (2026-05-30T08:39:39.004Z)

I have completed my verification and already submitted the StructuredOutput. Let me reconsider one important point before finalizing, because the WebSearch results actually surfaced evidence that materially affects my verdict.

The claim cites <https://www.sama.com/> as the source, but my fetches of the homepage and the video-annotation page found NO sports use case. However, WebSearch surfaced a real Sama page ? "Hitting Revenues Out of the Park with Accurate Data for Sports Analytics" (<https://www.sama.com/blog/data-labeling-challenges-for-sports-analytics>) ? plus keypoint-annotation and video-annotation service pages that DO reference sports analytics and player tracking. Let me verify these directly to confirm whether the claim's substance holds even if the exact cited URL is wrong.

21. user (2026-05-30T08:39:45.148Z)

Video Annotation Phrases - Verbatim Quotes

****Object tracking:****

"Our video annotation experts use bounding boxes to accurately segment videos used for a variety of applications including autonomous driving, food system optimization, and more. Bounding boxes are best used for object tracking and object detection."

****Temporal attributes:****

"Video Annotation: Object Tracking and Temporal Attributes" (appears in navigation menu describing the service)

****Linear interpolation between frames:****

"Utilize linear interpolation between frames" (appears multiple times across different annotation tool descriptions: bounding boxes, polygons, keypoints, lines & arrows, and cuboids sections)

****Static and dynamic annotations:****

"Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

****Complex object changes over time:****

"Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

****Sports analytics:****

"The latest paraphernalia assists us in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare."

****Player tracking:****

"The latest paraphernalia assists us in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare."

22. user (2026-05-30T08:39:46.500Z)

Sports Analytics and Motion Tracking from Sama's Keypoint Annotation

Yes, sports analytics is listed as a use case. Here's the exact sentence:

"Sports Analytics" appears under the Motion Tracking section in the USE CASES portion of the page.

Regarding motion tracking, here is the relevant quoted content:

"Motion Tracking" with associated applications including "Sports Analytics, Poses identification for AR/VR, Virtual fitness trainers"

The page indicates that keypoint annotation services support motion tracking applications across multiple domains, with sports analytics specifically mentioned as one practical implementation.

23. user (2026-05-30T08:39:47.181Z)

Based on the webpage content, Sama does describe video annotation capabilities relevant to sports. Here are the most relevant quotes:

****On video annotation services:****

"Object Tracking and Temporal Attributes" - this is listed under their Video Annotation service offering.

****On sports analytics application:****

"Camera feeds from previous matches and data on body angles when pitching or kicking collected from players' wearables all funnel into algorithms."

****On semantic labeling for player actions:****

"semantic information must be added in order to create context on what actions the players are performing: are they passing the ball, running with it, intercepting, defending, and so forth? How are their actions changing over time?"

The article emphasizes that Sama's platform handles the temporal dimension of sports video data?tracking how player actions and positions change across video frames?which is essential for comprehensive sports analytics.

24. user (2026-05-30T08:39:49.893Z)

Search Results for Specified Phrases

"linear interpolation between frames"

****Status:**** Appears verbatim multiple times.

****Examples:****

- "Utilize linear interpolation between frames"
- "Use ML-assisted or linear interpolation between frames"

"static and dynamic annotations"

****Status:**** Appears verbatim.

****Quote:**** "Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

"complex object changes over time"

****Status:**** Appears verbatim.

****Quote:**** "Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

"player tracking for sports analytics"

****Status:**** Does NOT appear verbatim.

****Closest equivalent:**** "The latest paraphernalia assists us in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare."

The phrase matches exactly within this sentence, so this phrase does appear verbatim in the document.

25. user (2026-05-30T08:40:10.577Z)

(Bash completed with no output)

26. user (2026-05-30T08:40:15.548Z)

Video Annotation Services Page Analysis

object tracking - "Bounding boxes are best used for object tracking and object detection."

temporal - "Utilize our video annotation service to revolutionize robotics and manufacturing. Our solution excels at identifying objects like parked and moving cars, even within your lane or during lane changes. We excel in pose identification?be it static, standing, sitting, and more. Additionally, our service can accurately detect emotional nuances and facial expressions, enhancing AI-driven systems across industries." (This appears under "Temporal Segmentation" heading)

interpolation - "Use ML-assisted or linear interpolation between frames" (appears multiple times for different annotation types)

static and dynamic - "Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

complex object changes - Same sentence as above: "Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations."

sports analytics - "The latest paraphernalia assists us in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare."

player tracking - Same sentence as above.

27. user (2026-05-30T08:40:46.079Z)

(Bash completed with no output)

28. user (2026-05-30T08:40:46.081Z)

Structured output provided successfully

29. user (2026-05-30T08:40:51.817Z)

Verbatim Sentences with Target Words

***Object tracking**: **

- "Bounding boxes are best used for object tracking and object detection."
- "Use ML Assist for faster high-quality annotations" (in context of object tracking)
- "Leverage our comprehensive video annotation service to revolutionize multiple sectors. In autonomous check-out scenarios, we ensure precise item identification for seamless transactions. Our expertise extends to annotating furniture, enabling immersive AR/VR experiences. For cutting-edge autonomous vehicles, we specialize in accurate vehicle and pedestrian tracking, enhancing safety and navigation."

***Temporal**: **

- "Utilize our video annotation service to revolutionize robotics and manufacturing. Our solution excels at identifying objects like parked and moving cars, even within your lane or during lane changes."

***Interpolation**: **

- "Use ML-assisted or linear interpolation between frames"
- "Utilize linear interpolation between frames" (appears three times in polygon, keypoint, and lines/arrows sections)

***Static and dynamic**: **

- "Improve your ML model's accuracy by tracking complex object changes over time with static and

dynamic annotations."

***Sports:**

- "The latest paraphernalia assists us in player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare."

Named Use Cases

1. Spatial Object Tracking
2. Harvest Yields With Computer Vision
3. Temporal Segmentation

30. assistant (2026-05-30T08:41:00.168Z)

The deeper verification confirmed my corrected verdict. The direct WebFetch of the correct Sama page (<https://www.sama.com/2d-video-annotation-tool-services>) now succeeded and verified every quoted phrase verbatim:

- "Utilize linear interpolation between frames" / "Use ML-assisted or linear interpolation between frames" ? appears multiple times
- "Improve your ML model's accuracy by tracking complex object changes over time with static and dynamic annotations" ? verbatim
- "Object Tracking and Temporal Attributes" ? verbatim service heading; a "Temporal Segmentation" use case section also exists
- "...player tracking for sports analytics, object recognition for autonomous vehicles, and medical image analysis in healthcare" ? verbatim

The keypoint-annotation page independently lists "Sports Analytics" under Motion Tracking use cases, and Sama publishes a dedicated sports-analytics blog post.

My final StructuredOutput (refuted=false, medium confidence) stands. The claim's substance is accurate and primary-source-verified. The one defect ? which I flagged for the report author ? is that the claim cites the homepage URL (<https://www.sama.com/>), where these phrases do NOT appear; the correct citation is <https://www.sama.com/2d-video-annotation-tool-services>. The claim's own caveat that Sama markets sports tracking "though no detailed sports portfolio is shown" is fair: the sports references are marketing copy and a generic blog, with no concrete sports case study.